

Notes of the Genius Society

DISCLAIMER: This is a *Theorycraft*. We are constantly testing, experimenting, and updating the method to upgrade. The method nor outcome is set in stone

NOTE: This **does not involve cheating/hacking** the game. This is just a Data Analysis project, where large quantities of relic upgrades are recorded and studied for patterns. It's using what's in the game to the player's advantage.

THE MAIN GOAL OF THIS SPREADSHEET IS TO HELP PLAYERS ROLL BETTER

Recommended Resources

✚ [HSR string recording Template](#)

✚ [Notes of the Genius Society](#)

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Section 1 - Background Theory

General description of theory

‣ This theory suggests that there are a long "string" of numbers like 12131121423313341112131421 etc which determines the pattern of your upgrades. The way the "RNG" works is it starts you off on a "random" starting point. Every time you do an upgrade, you move one spot further on the RNG string, but also after a certain amount of time (15minutes) it moves down the string automatically or repositions you to another point on the string based on the second layer of pseudo-RNG. The Numbers determine what line of the artifact is being upgraded.

‣The way that we make use of this, is by recording and building up the data for the upgrades for the different times. The findings of these recordings is that there are certain times where some strings are more common and less common. When there is enough data, other people can base their upgrades on this data. You can also base your upgrades on your own previous data within the same 15 minute session.
In a way we are tracking the movement of the upgrades from your starting point.

How to read strings

When we read the strings in our data we separate it into 5 numbers. This represents the pattern of the upgrades in chronological order from left to right. Each number represents what line in the substat you are upgrading. When a number is unknown in the string, then it is represented as x. [Please refer to [Example 1](#)].

Example 1:

1 = Substat Line 1

2 = Substat Line 2

3 = Substat Line 3

4 = Substat Line 4



How to create/reset strings

There are a few things that will start/reset a string:

- Adding a substat will reset and start a new string on the line you just added.

(Main method used) [[Example 2](#)]

- Continually upgrading without resetting means that the 5th number in the string will become the first number of the next string (Random. Not recommended to do this)

[[Example 3](#)]

- Going into a 4 substat relic and upgrading without doing anything prior will mean that whatever line gets upgraded is the first number of the string (only if it's your first upgrade of a session). (Yields a random starting number. Not recommended to do this but can be used if you don't have low tier relics to start 1xxxx or 2xxxx strings) [[Example 4](#)]

▸ Note that exiting the character screen will make you lose progress on your strings

▸ In this method, relic strings carry over and do not reset across relics. **If you continue upgrading, then it will continue the string as long as you don't trigger a reset.**

<u>Example 2.1</u>		
lvl 0	`+3	lvl 3
Relic/Artifact	->	Relic/Artifact
Substat 1		Substat 1
Substat 2		Substat 2
Substat 3		Substat 3
		Substat 4
As you added line 4, String is now 4xxxx		

<u>Example 2.2</u>		
lvl 9	`+3	lvl 12
Relic/Artifact	->	Relic/Artifact
Substat 1		Substat 1
Substat 2		Substat 2
		Substat 3
If you had a previous string, it is gone now.		
As you added line 3, String is now 3xxxx		

<u>Example 3</u>		
Combined		

string		
41423343314	->	41423
		33433
		314xx
Remember that the 5th number in the upgrade rarely has data so it will be random/unpredictable		

Example 4		
lvl 0	`+3	lvl 3
Relic/Artifact	->	Relic/Artifact
Substat 1		Substat 1
Substat 2		Substat 2
Substat 3		Substat 3
Substat 4		Substat 4
As you upgraded line 2. String is now 2xxxx		

How to use translation

- The strings actually represent a pattern which shows the movement/direction where the lines are upgraded in relation to the starting string.
- The string 4132x also means x, x-3, x-1, x-2. Where x is represented as the starting position of the string. In this case, the starting number is 4.
- You can also visualise 4132x as x-3+2-1

- When your resulting pattern adds over the 4th line, it loops back to the first line. If the pattern subtracts less than 1, it loops back into 4. You can imagine the number line like ...123412341234.... [Example 5]
- When you look at the main sheet, It will show a percentage for a certain string, for example 444 has a 40% chance to hit. When you go to your relic and your crit line is on line 2, you can utilise translation. 444 has a pattern of x, x+0, x+0, x+0 so if you start a string on 2xxxx, then you will hit 222xx pattern on the relic instead. Refer to [Example 6] for another example situation.
- Remember that your first upgrade should be forcing a new string to start on the line you want and not on your actual relic. The second upgrade (the second number in the string) and onwards is where you switch to the relic you want to upgrade.

Example 5:

If An Upgrade Steps Up from substat Line 1 then it hits substat line 4



If An upgrade steps Down from substat Line 4 then it hits Substat line 1

<u>Example 6</u>		
lvl 3	Practice situation	

Relic/Artifact	Assuming that it was guaranteed that you will land any of the top 2 highest chance outcomes in the next upgrade, how should you translate?
ATK	
Crit Rate	
Crit DMG	
Effect RES	

Referencing the summary sheet		
		Can you tell me the 2 highest string chances? Type your answers below

String pattern identification		
44 =		What formula are the strings?
41 =		
Type your answers in the ^ above box		
Answers		
To check your answers highlight or change the colour of the answer cells.		

What Should you translate the string to?
--

If your aim is to hit any of the crit substats, which are in line 2 or 3, what line should you start a string on to make those the only possibilities?

Hint Your starting string is x

^ Type your answers in the above box

Answer

To check your answers highlight or change the colour of the answer cells.

Congratulations

lvl 6	Perfect scenario	
Relic/Artifact	You hit the second most likely outcome of 41. You were able to make it land into line 3 instead of line 1 by using translation. Nice	
ATK		
Crit Rate		
Crit DMG +1		
Effect RES		

Section 2 - Tools available

Time theory explanation

- The current theory states that the strings are time sensitive to the patch/day/15mins.
- The data within patches will change every patch - Every individual day will be the same each week (E.g. Each monday has the same data).
- Every 15mins is the most consistent time to upgrade without worrying about pattern changes.
- Longer strings seem to maintain consistency over longer periods of time (half hour/hour/day). This is mostly due to how longer strings work.
- The old method of aiming for longer strings across the whole day would still work the same as before (if you desire to) but it should be more accurate to work with data (both the sheet and your own) within 15 minutes.

How to navigate the String Data Sheet

- The new spreadsheet will look like this.

A	B	C	D	E	F	G	H	I	J	K	L
	Patch	Server		Day							
	2.1	America		Saturday		Choose formula for auto selection to today every time!					
Timeframe selection:											
15	12:00 AM to 12:15 AM	12:15 AM to 12:30 AM	12:30 AM to 12:45 AM	12:45 AM to 1:00 AM	1:00 AM to 1:15 AM	1:15 AM to 1:30 AM	1:30 AM to 1:45 AM	1:45 AM to 2:00 AM	2:00 AM to 2:15 AM	2:15 AM to 2:30 AM	2:30 AM to 2:45 AM
42	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	50.00%	0.00%	0.00%	40.00%
43	0.00%	0.00%	0.00%	0.00%	0.00%	100.00%	0.00%	50.00%	44.44%	0.00%	6.67%
44	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	22.22%	0.00%	13.33%
Count	0	0	0	0	1	1	0	2	9	0	15

The sheet will show you data that is popular for each timeframe that is selected. Select the timeframe you want and then try to hit that data. There are different uses for the different time frames but 15mins is the recommended one.

- Settings to note:

1. Make sure that your server is set to your ingame server.
2. Make sure that your day is set to the day you want OR choose the option for automatic day update
3. For those of you who worry about a laggy sheet, you can disable data for each section of the longer strings
4. **You do not want to change the sheet timezone settings.** All you need to look at is the **current sheet time** and you can look at the data for your corresponding times.

I	J	K	L	M	N
				Current sheet time	
				1:36:37 PM	

+

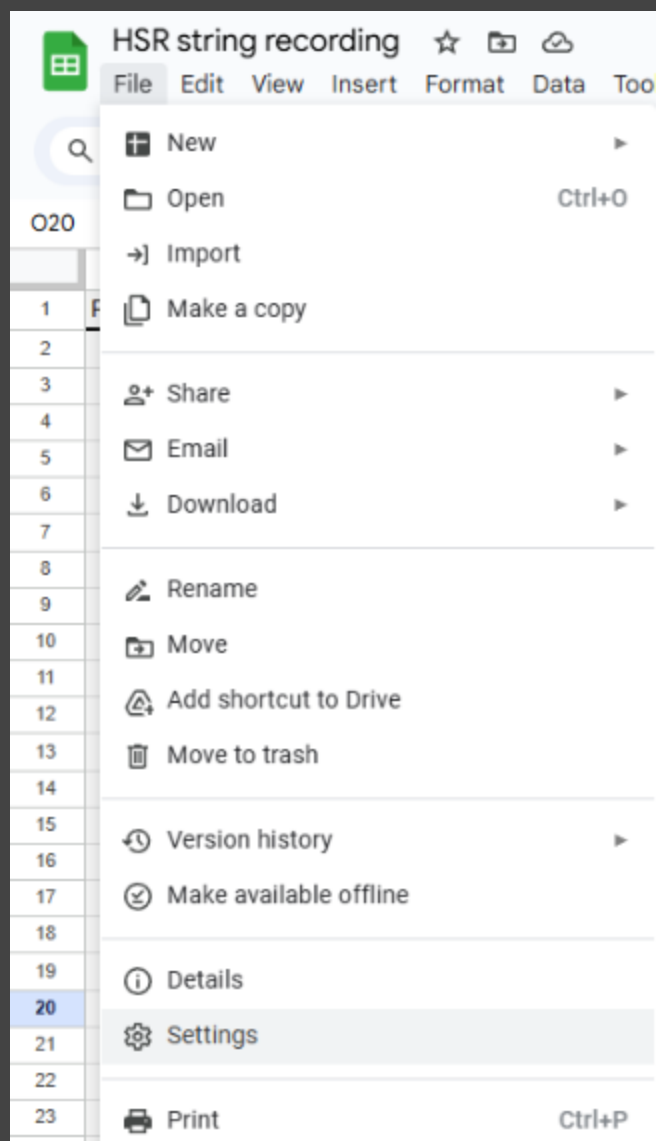
1:30 AM to 1:45 AM
0.00%
0.00%
0.00%
0.00%
0

How to Use the String Recording Sheet

Recording

The basic function of the string recording sheet will allow for you to record all your strings and assign approximate times to them. This will be very important for data submission as we will not take any data without times.

- The next main function of this sheet is to provide a variety of tools to help you recognise any patterns in your data and help with predicting what your rolls will be like.
- To begin using the sheet, there are a few things you need to adjust first.
- Make sure you change the region of your sheet to your own, or else the times that are recorded wont be the same time as your own.



- type the strings that you have rolled into the RAW ROLLS column and the rest of the information will automatically fill out for you.

Count	Raw Rolls	Translated	Time	2 String	3 string	4 string	5 string
1	412	41200	12:00:00 AM	41	412		
2	41	41000	4:44:12 PM	41			
3	4121	41210	4:44:12 PM	41	412	4121	
1	444	44400	4:44:12 PM	44	444		
2	44	44000	4:44:12 PM	44			
3	444	44400	4:44:12 PM	44	444		
1	421	42100	4:44:12 PM	42	421		
2	43	43000	4:44:12 PM	43			
3	4423	44230	4:44:12 PM	44	442	4423	
1	434	43400	4:44:12 PM	43	434		
2	41	41000	4:44:12 PM	41			
3	4311	43110	4:44:12 PM	43	431	4311	
4	43	43000	4:44:12 PM	43			
5	412	41200	4:44:12 PM	41	412		
1	441	44100	4:44:12 PM	44	441		
2	43	43000	4:44:12 PM	43			
3	4341	43410	4:44:12 PM	43	434	4341	
4	44	44000	4:44:12 PM	44			
5	42	42000	4:44:12 PM	42			
6	412	41200	4:44:12 PM	41	412		

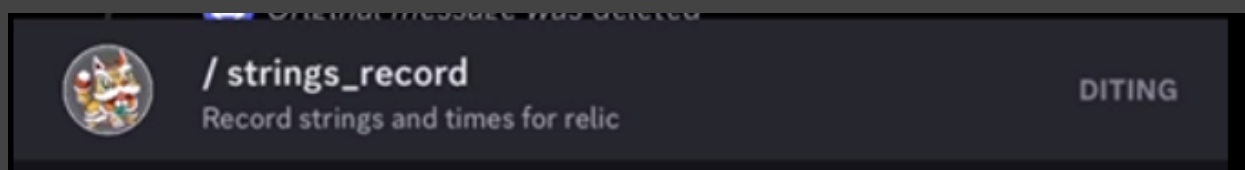
Submitting data

When you submit data, just copy both your translated column and time column

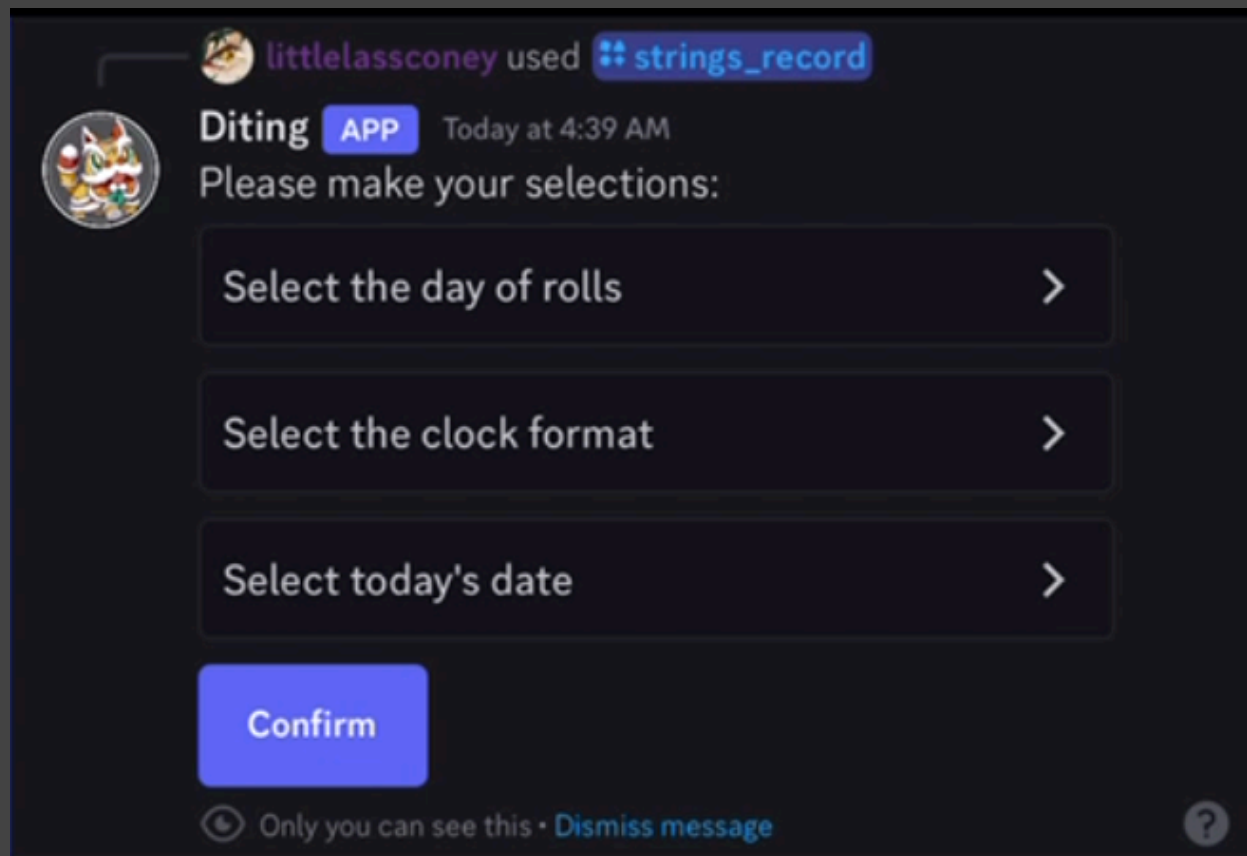
Count	Raw Rolls	Translated	Time	2 String
1	44	44000	4:44:44 PM	44
2	41	41000	4:44:46 PM	41
3	42	42000	4:44:47 PM	42
4	42	42000	4:44:48 PM	42
5	44	44000	4:44:48 PM	44
6	44	44000	4:44:49 PM	44
7	42	42000	4:44:50 PM	42
8	43	43000	4:44:51 PM	43
9	42	42000	4:44:52 PM	42
10	41	41000	4:44:52 PM	41
11	42	42000	4:44:52 PM	42
12	42	42000	4:44:53 PM	42
13	43	43000	4:44:53 PM	43
14	41	41000	4:44:54 PM	41
15	41	41000	4:44:55 PM	41
16	43	43000	4:44:56 PM	43
17	41	41000	4:44:56 PM	41
18	41	41000	4:44:56 PM	41
19	44	44000	4:45:30 PM	44
20	41	41000	4:45:31 PM	41
21	43	43000	4:45:32 PM	43
22	41	41000	4:45:32 PM	41
23	43	43000	4:45:33 PM	43
24	41	41000	4:45:33 PM	41

Once you have your data, head over to our discord and make sure you go to the correct server data channel that you are playing on.

For example if you are on asia server, go to # asia-pattern-data-hsr.



1.Type /Strings_record to bring up the menu



2. Fill out your day, clock format and current time/date.



Strings record



This form will be submitted to Diting. Do not share passwords or other sensitive information.

Current Time *

4:40 am

Strings & Times *

41 4:40am

3989

Submit

3. Paste your translated rolls and times into the box and press submit.
Your data will now be verified before being added into the data pool.

Notes column

This is just something that could be useful in showing your predictions or keeping track of them. The cell colour just goes red if you type wrong anywhere.

I	J	K	L
	Predictions/notes	No.	
	green with any text		
	red if has "wrong"	1	
	i think 44 next		
		2	
	i think 42 next wrong		
	44 next		
		3	
	41 next		
	43		
	41 wrong		
		4	

New session and string rate

Whenever you make a space in the string recording sheet, it is treated as a new session, removing the need to continually making new sheets or clearing them out. Just make a space and add under.

1	4224	42240	11:32:25 PM	42	422	4224	
2	41	41000	11:34:14 PM	41			
3	422	42200	11:35:59 PM	42	422		
4	44	44000	11:38:24 PM	44			
1	4412	44120	12:07:25 AM	44	441	4412	
2	43	43000	12:08:17 AM	43			
3	41	41000	12:09:13 AM	41			
4	42	42000	12:11:44 AM	42			
1	411	41100	12:17:15 AM	41	411		
2	42	42000	12:18:21 AM	42			
3	4333	43330	12:19:00 AM	43	433	4333	
4	41	41000	12:22:27 AM	41			
5	41	41000	12:24:04 AM	41			
6	42	42000	12:25:16 AM	42			
7	43	43000	12:26:21 AM	43			
1	432	43200	12:32:37 AM	43	432		
2	41	41000	12:33:27 AM	41			
3	4312	43120	12:35:15 AM	43	431	4312	
4	41	41000	12:36:01 AM	41			
5	43	43000	12:37:10 AM	43			
6	44	44000	12:38:04 AM	44			
7	43	43000	12:39:18 AM	43			

Each of your sessions will be numbered in the No. column and will also provide the 4x percentages for that whole session. So if you want to test distributions for different lengths of time, just make a space at what you want.

No.	41/42	41/43	41/44	41	42	43	44
	42	42	42				
	41	41	41				
	42	42	42				
1	44	44	44	25%	50%	0%	25%
	44	44	44				
	43	43	43				
	41	41	41				
2	42	42	42	25%	25%	25%	25%
	41	41	41				
	42	42	42				
	43	43	43				
	41	41	41				
	41	41	41				
	42	42	42				
3	43	43	43	43%	29%	29%	0%
	43	43	43				
	41	41	41				
	43	43	43				
	41	41	41				
	43	43	43				
	44	44	44				
4	43	43	43	29%	0%	57%	14%

Possible groupings - 4x

This function takes your current rolls and colours them based on all possible combinations for 4x patterns. This way it will be easier to account for your pattern being different from others as well as give you a better visual idea on the pattern (so you can try to predict/make use of it). The common use of it would be to see what is common for the time period that you are upgrading in, for example in the example image it goes from 41/42 to 44/43 to 41/42 to 43/41 to 43/42. It can be easier to see

L	M	N	O	P
	41/42	41/43	41/44	
	42	42	42	
	41	41	41	
	42	42	42	
	44	44	44	
	44	44	44	
	43	43	43	
	41	41	41	
	42	42	42	
	41	41	41	
	42	42	42	
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	43	43	43	
	44	44	44	
	43	43	43	
	43	43	43	
	42	42	42	
	43	43	43	
	43	43	43	
	41	41	41	
	42	42	42	
	43	43	43	
	41	41	41	

Upcoming features

Highest consecutive occurrence table

This just keeps track of the highest times in a row that each group takes in your current data.

Formula that tips you off when pity has been hit

Long string patterns

String recording program

Something that records the time more precisely when you do your upgrades.

Section 3 - Upgrading Method

Most basic method

The most Common upgrading method is to look on the data summary sheet and observe the most common roll, and roll for it without regard for anything else. You will have less control when doing it this way and success would be more reliant on having the popular lines be extremely dominant at that specific time (it is a valid method).

We can go through an example here:

15	10:00 PM to 10:15 PM	1
41	17.50%	
42	35.00%	
43	25.00%	
44	22.50%	
Count	40	

The data shows that between 10:00-10:15pm there is the highest chances to hit 42 and 43 (combined 70% chance to hit) so within this session, you will mainly try to hit those two lines in particular.

1. Start a 4xxxx string on any artifact and then swap to the relic you want to upgrade to continue the string. Make sure that 42 and 43 outcomes are both beneficial for the artifact for best measure (e.g crit on line 2 and 3)

assume you hit 42 in this scenario

If you want to continue the upgrade string, then look at the section with 42xxx strings and look at the rates.

421	42.86%
422	28.57%
423	28.57%
424	0.00%
Count	7

This is usually where you can start to see greater favouring towards certain outcomes and this is usually why you go for longer strings. In this image, you can see that 421xx occurs 42% of the time. Keep in mind to pay attention to the count for the data as if it is lower, the data is less reliable and vice versa. In general, you may see

that longer strings will tend to favour certain strings more so you can look out for that in your own data as well if you are upgrading without the sheet.

2. Assuming you want to hit this data, make sure you do not accidentally reset the string. Your current string is 42xxx so you can aim for a good substat in the first line. If your artifact has crit on lines 2 and 3 but line 1 has hp, then swap to another 4 substat artifact with a good stat on line 1 to do the upgrade.

Predicting patterns

When you are looking at your data, it is unreasonable to think that it will always roll into the most popular lines, therefore you should always incorporate predicting when it will swap if possible. Remember that the strings patterns will last for 15mins so thats how long you can build data for your predictions. If there is a pattern that lasts for longer than 15mins, it is likely just a similar back to back pattern.

There are a few variables that you should take into account when looking at your rolls.

- What are the popular strings in your session. The the first line to show up is always the most popular roll in your session.
- How many times in a row does each group usually hit before switching. Its safer to base your prediction on the maximum possible amount. You can also give higher confidence to that number the more times it happens.
- The speed at which you upgrade should be consistent if you can do it. The patterns seem to switch with either "enough time has passed" or "you have hit that group too many times already". (This could be wrong or right but needs more testing.)

The Method BBP uses to upgrade relics is by going for long string biased data and taking into account these variables.

Doing test strings

Doing test strings are important in order to check what your data is looking like for the current session. They shouldnt be based on a specific amount but only to determine the pattern or find out the information that you need in order to upgrade. Doing longer lines gives you a better idea of what your session looks like (can do less) and doing shorter strings lets you do more of them (but they provide less data). Another note is that the distribution of lines in the length of the same string seems to show the distribution of the lines in the upcoming new strings. For example if you get 43324, you can tell that in the upcoming strings that 43xxx will be popular and 41xxx will be the least popular.

The current best method

The current best way to do upgrades is to just upgrade within 15minutes using your own data, while keeping in mind all characteristics/variables in the data. Try to think for yourself and test out things. The tools are there so please try some testing/creating your own things and share it with the community. I have refrained from writing a method that is too specific for general use as there is not one that I can confidently recommend that will give you the best rolls all of the time. There are other methods i might plan to write but i am too busy for now.

Distribution method (Esme/Whiscat)

This method is more focused on 4x string data rather than using long strings. Tldr uses long strings to predict the least likely line and avoid that in the first few rolls.

- *It will do guide if requested*

Matching data Method (Kiyoko)

- *Same as above*
-

Table of tips

I'll try to make a list of all the tips and rules that all data seems to follow. These come from the people who are more experienced with rolling and hopefully helps the newer people.

Happens most/all the time	Observations (needs testing)
- Your 4x strings change up every 15mins. Sometimes you can get similar ones but almost every time (80-90%) its different. - The first line to show up is always the most popular roll in your session. -	- The distribution of lines in the length of the same string seems to show the distribution of the lines in the upcoming new strings. For example if you get 43324, you can tell that in the upcoming strings that 43xxx will be popular and 41xxx will be the least popular. -

